

CSEC Mathematics
June 2018 – Paper 2
Solutions

SECTION I

Answer ALL questions.

All working must be clearly shown.

1. (a) Using a calculator, or otherwise, evaluate EACH of the following, giving your answer to two decimal places.

(i) $73.18 - 5.23 \times 9.34$ [1]

$$\begin{aligned} & 73.18 - 5.23 \times 9.34 \\ &= 73.18 - (5.23 \times 9.34) \\ &= 73.18 - 48.8482 \\ &= 24.3318 \\ &= 24.33 \text{ (to 2 decimal places)} \end{aligned}$$

(ii) $\frac{3.1^2}{6.17} + 1.12$ [1]

$$\begin{aligned} & \frac{3.1^2}{6.17} + 1.12 \\ &= \frac{9.61}{6.17} + 1.12 \\ &= 1.5575 + 1.12 \\ &= 2.6775 \\ &= 2.68 \text{ (to 2 decimal places)} \end{aligned}$$

(b) Jenny works at Sammy's Restaurant and is paid according to the rates in the following table.

Jenny's weekly wage agreement
Basic wage \$600.00 PLUS \$0.90 for each customer served

In a week when Jenny serves n customers, her weekly wage, W_j , in dollars, is given by the formula

$$W_j = 600 + 0.90n$$

- (i) Determine Jenny's weekly wage if she served 230 customers. [1]

$$W_j = 600 + 0.90n$$

$$\text{When } n = 230,$$

$$W_j = 600 + (0.9 \times 230)$$

$$W_j = 600 + 207$$

$$W_j = 807$$

$\therefore$ Jenny's weekly wage is \$807 if she serves 230 customers.

- (ii) In a **good** week, Jenny's wage is \$1 000.00 or more. What is the LEAST number of customers that Jenny must serve in order to have a **good** week? [2]

$$W_j = 600 + 0.9n \quad \text{and} \quad W_j \geq 1000$$

Hence,

$$600 + 0.9n \geq 1000$$

$$0.9n \geq 1000 - 600$$

$$0.9n \geq 400$$

$$n \geq \frac{400}{0.9}$$

$$n \geq 444.4$$

We need to round up in order for Jenny to earn at least \$1000.

$\therefore$ The least number of customers Jenny must serve is 445.

- (iii) At the same restaurant, Shawna is paid a weekly wage of \$270.00 plus \$1.50 for each customer she serves.

If W_s is Shawna's weekly wage, in dollars, write a formula for calculating Shawna's weekly wage when she serves m customers. [1]

$$W_s = \text{Shawna's salary} \quad \text{and} \quad m = \text{number of customers}$$

$$\therefore W_s = 270 + 1.5m$$

- (iv) In a certain week, Jenny and Shawna received the same wage for serving the same number of customers.

How many customers did they EACH serve? [3]

$$W_j = 600 + 0.9n \quad \text{and} \quad W_s = 270 + 1.5m$$

If the number of customers served by both Jenny and Shawna is the same, then $n = m$.

Then,

$$600 + 0.9n = 270 + 1.5n$$

$$600 - 270 = 1.5n - 0.9n$$

$$330 = 0.6n$$

$$n = \frac{330}{0.6}$$

$$n = 550$$

$\therefore$ Jenny and Shawn each served 550 customers.

Total: 9 marks

2. (a) Factorize, completely, EACH of the following expressions.

(i) $1 - 4h^2$ [1]

$$\begin{aligned} &1 - 4h^2 \\ &= (1)^2 - (2h)^2 \quad \text{[difference of two squares]} \\ &= (1 - 2h)(1 + 2h) \end{aligned}$$

(ii) $pq - q^2 - 3p + 3q$ [2]

$$\begin{aligned} &pq - q^2 - 3p + 3q \\ &= q(p - q) - 3(p - q) \\ &= (q - 3)(p - q) \end{aligned}$$

(b) Solve EACH of the following equations.

(i) $\frac{3}{2}y = 12$ [1]

$$\begin{aligned} &\frac{3}{2}y = 12 \\ &3y = 12 \times 2 \\ &3y = 24 \\ &y = \frac{24}{3} \\ &y = 8 \end{aligned}$$

(ii) $2x^2 + 5x - 3 = 0$

$$2x^2 + 5x - 3 = 0$$

$$2x^2 + 6x - x - 3 = 0$$

$$2x(x + 3) - 1(x + 3) = 0$$

$$(2x - 1)(x + 3) = 0$$

Either $2x - 1 = 0$ or $x + 3 = 0$

$$2x = 1$$

$$x = -3$$

$$x = \frac{1}{2}$$

(c) The quantities F, m, u, v and t are related according to the formula

$$F = \frac{m(v-u)}{t}$$

(i) Find the value of F when $m = 3, u = -1, v = 2$ and $t = 1$. [1]

$$F = \frac{m(v-u)}{t}$$

When $m = 3, u = -1, v = 2$ and $t = 1$

$$F = \frac{3(2-(-1))}{1}$$

$$= 3(2 + 1)$$

$$= 3(3)$$

$$= 9$$

∴ The value of $F = 9$.

(ii) Make v the subject of the formula.

[2]

$$F = \frac{m(v-u)}{t}$$

$$\frac{F}{1} = \frac{m(v-u)}{t}$$

Cross-multiplying gives:

$$Ft = m(v - u)$$

$$Ft = mv - mu$$

$$Ft + mu = mv$$

$$mv = Ft + mu$$

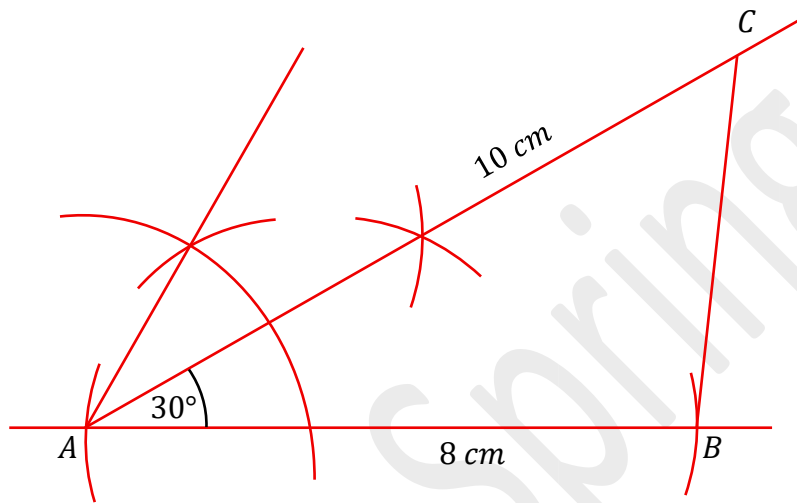
$$v = \frac{Ft + mu}{m}$$

Total: 9 marks

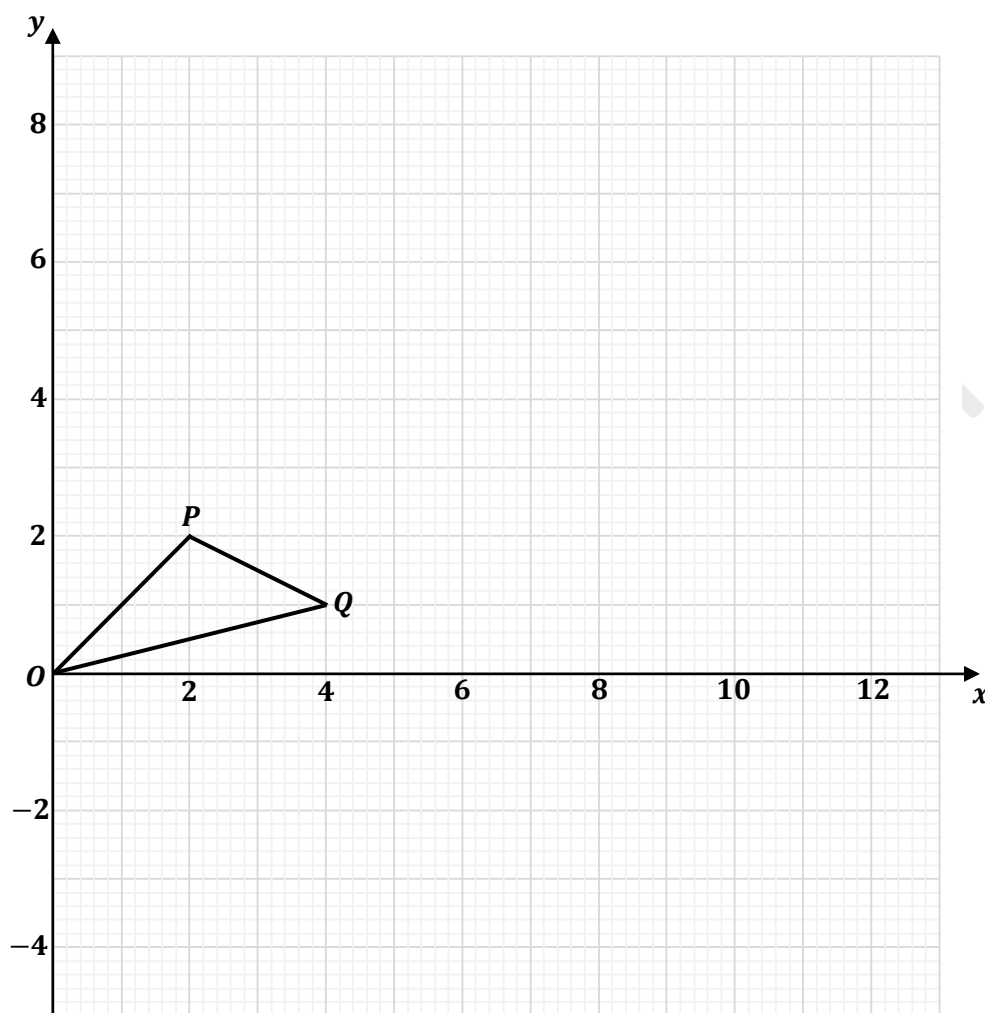
3. (a) Using a ruler, a pencil and a pair of compasses, construct the triangle ABC ,

such that $AB = 8\text{ cm}$, $\angle BAC = 30^\circ$ and $AC = 10\text{ cm}$.

[4]



(b) The diagram below shows the triangle OPQ .



- (i) State the coordinates of the point Q . [1]

The coordinate of the point Q is $(4, 1)$.

- (ii) The line PQ is mapped to $P'Q'$ by an enlargement, centre O and scale factor 3. On the diagram above, draw the line $P'Q'$. [2]

$$P = (2, 2)$$

Now,

$$\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 6 \end{pmatrix}$$

$$\therefore P' = (6, 6)$$

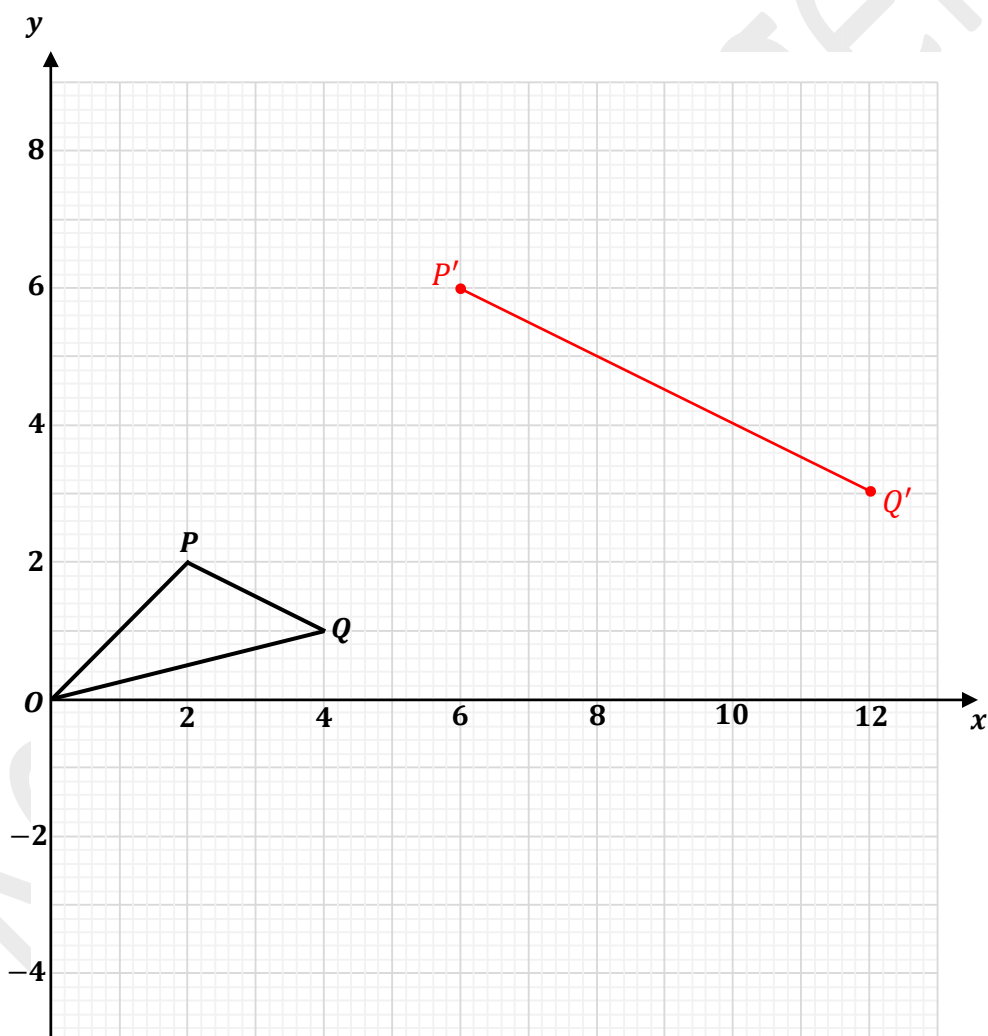
$$Q = (4, 1)$$

Now,

$$\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 12 \\ 3 \end{pmatrix}$$

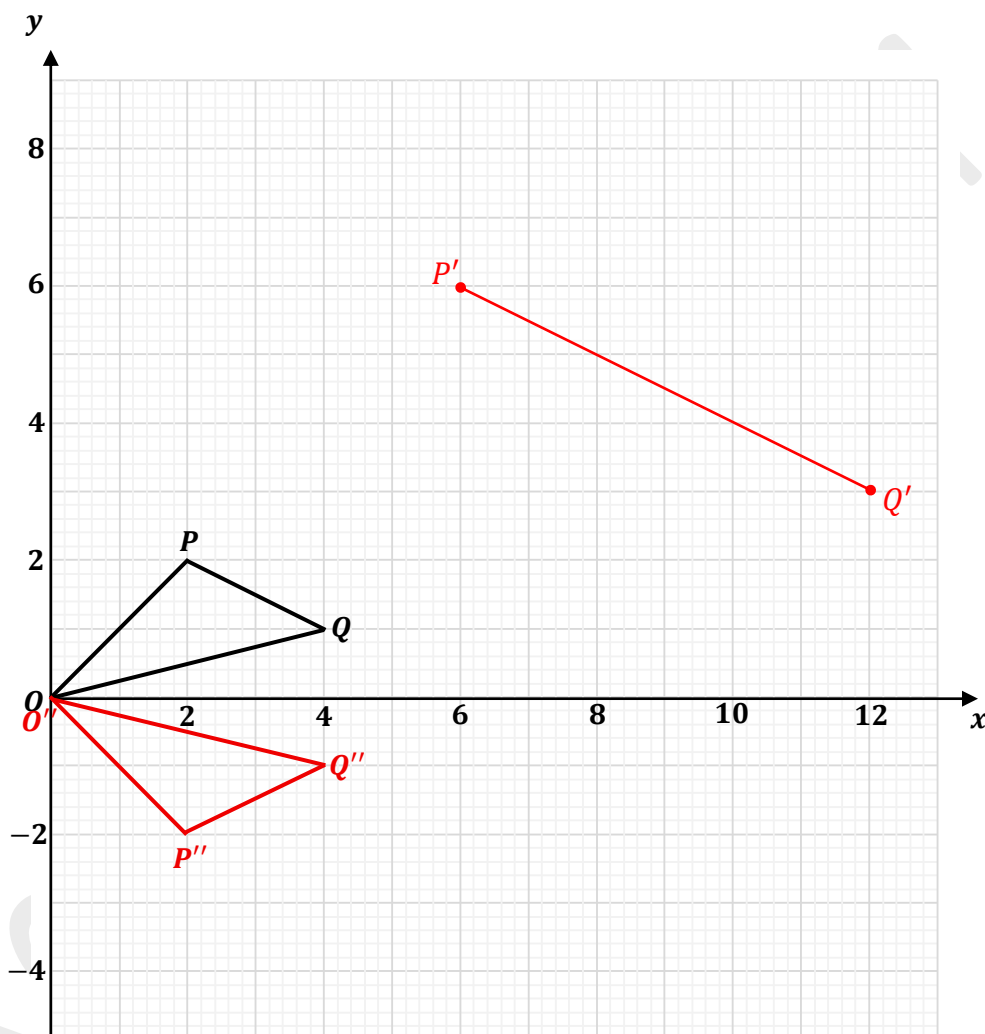
$$\therefore Q' = (12, 3)$$

The line $P'Q'$ is shown below:



- (iii) The $\triangle OPQ$ undergoes a reflection in the line $y = 0$ to produce the image $O''P''Q''$. On the diagram above, draw the $\triangle O''P''Q''$. [2]

The line $P'Q'$ is shown below:



Total: 9 marks

4. (a) The function f with domain, $A = \{1, 2, 3\}$ is given by

$$f(x) = \frac{1}{2}x - 3.$$

- (i) What is the value of $f(1)$? [1]

$$f(x) = \frac{1}{2}x - 3$$

When $x = 1$,

$$f(1) = \frac{1}{2}(1) - 3$$

$$= \frac{1}{2} - 3$$

$$= -\frac{5}{2}$$

- (ii) Find the value of x for which $f(x) = -2$. [1]

$$f(x) = -2$$

So, we have,

$$\frac{1}{2}x - 3 = -2$$

$$\frac{1}{2}x = -2 + 3$$

$$\frac{1}{2}x = 1$$

$$x = 1 \times 2$$

$$x = 2$$

- (iii) An ordered pair for the function is expressed in the form (a, b) . Using your answers to (a)(i) and (a)(ii), or otherwise, list the ordered pairs for the function, f . [2]

From (a)(i), $f(1) = -\frac{5}{2}$

From (a)(ii), $f(2) = -2$

$$\begin{aligned} f(3) &= \frac{1}{2}(3) - 3 \\ &= \frac{3}{2} - 3 \\ &= -\frac{3}{2} \end{aligned}$$

So, the list of ordered pairs is $\left\{\left(1, -\frac{5}{2}\right), (2, -2), \left(3, -\frac{3}{2}\right)\right\}$.

- (iv) Explain why $f(x) \neq 5$ for the function specified on page 13. [1]

If $f(x) = 5$, then

$$\frac{1}{2}x - 3 = 5$$

$$\frac{1}{2}x = 5 + 3$$

$$\frac{1}{2}x = 8$$

$$x = 8 \times 2$$

$$x = 16$$

But 16 is out of the domain, A , of the function which includes $\{1, 2, 3\}$.

$\therefore f(x)$ cannot be equal to 5.

(b) (i) Solve the inequalities

$$(a) 3x - 1 < 11 \quad [1]$$

$$3x - 1 < 11$$

$$3x < 11 + 1$$

$$3x < 12$$

$$x < \frac{12}{3}$$

$$x < 4$$

$$(b) 2 \leq 3x - 1 \quad [1]$$

$$2 \leq 3x - 1$$

$$2 + 1 \leq 3x$$

$$3 \leq 3x$$

$$\frac{3}{3} \leq x$$

$$1 \leq x$$

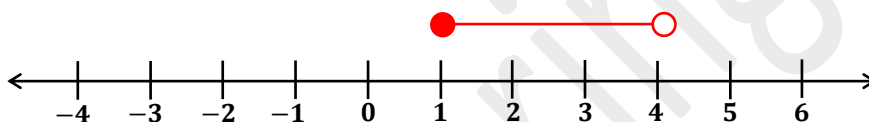
$$x \geq 1$$

- (ii) Represent the solution to $2 \leq 3x - 1 < 11$ on the number line shown below. [2]

The two inequalities are $x < 4$ and $x \geq 1$.

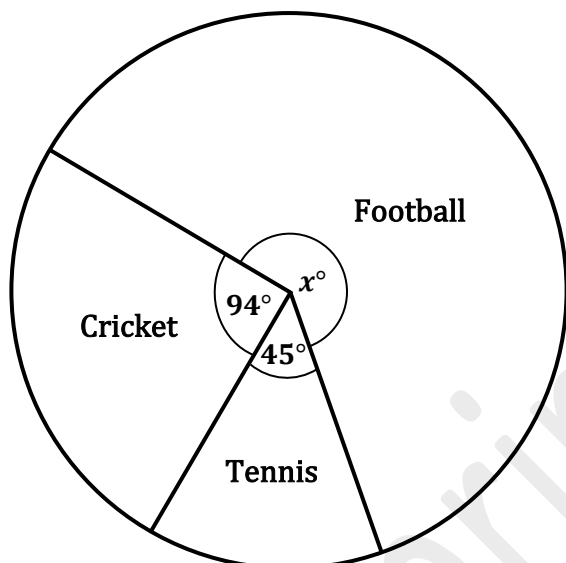
The solution to $2 \leq 3x - 1 < 11$ is $1 \leq x < 4$.

The inequality $1 \leq x < 4$ is represented on the number line below:



Total: 9 marks

5. (a) Students in a group were asked to name their favourite sport. Their responses are shown on the pie chart below.



- (i) Calculate the value of x . [1]

The sum of the angles in a circle add up to 360° .

$$\begin{aligned} x &= 360^\circ - (94^\circ + 45^\circ) \\ &= 221^\circ \end{aligned}$$

- (ii) What percentage of the students chose cricket? [1]

The percentage of students who chose cricket

$$= \frac{\text{Angle corresponding to cricket}}{360^\circ} \times 100\%$$

$$= \frac{94^\circ}{360^\circ} \times 100\%$$

$$= 26.11\% \quad (\text{to 2 decimal places})$$

- (iii) Given that 40 students chose tennis, calculate the TOTAL number of students in the group. [2]

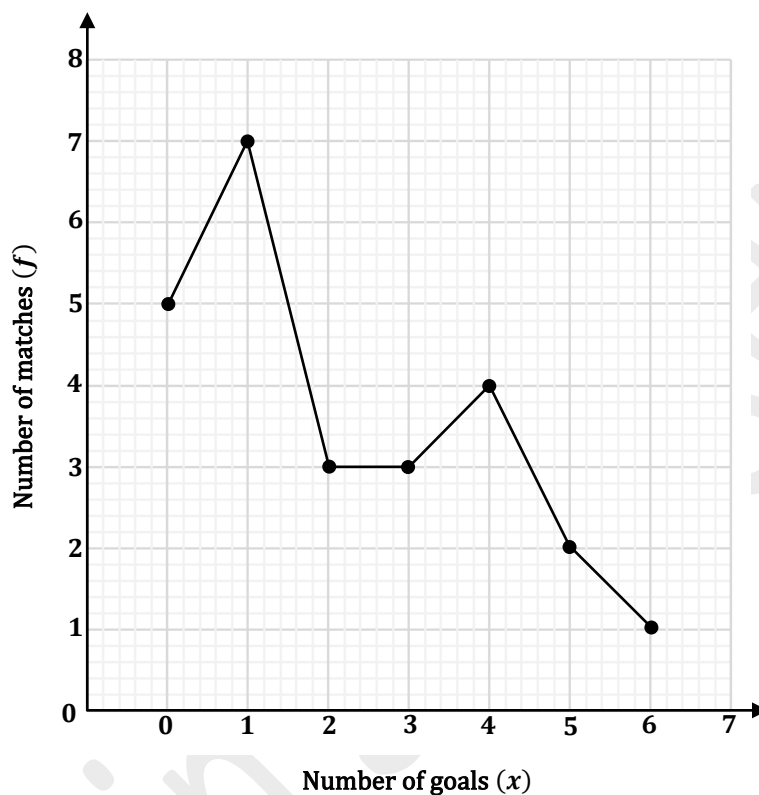
$$\begin{aligned} \text{Percentage of students who chose Tennis} &= \frac{45^\circ}{360^\circ} \times 100\% \\ &= 12.5\% \end{aligned}$$

If 12.5% of students equals 40 students, then

$$\begin{aligned} 100\% \text{ of students} &= \frac{40}{12.5} \times 100 \\ &= 320 \end{aligned}$$

$\therefore$ The total number of students is 320.

- (b) The diagram below shows a frequency polygon of the number of goals scored by a football team in 25 matches.



- (i) Complete the following table using the information in the diagram.

[1]

Number of matches (f)	5	7	3	3	4	2	1
Number of goals scored (x)	0	1	2	3	4	5	6

- (ii) What is the modal number of goals scored by the team?

[1]

The team scored 1 goal in 7 matches, this score occurs the most, having the highest frequency. Therefore, the modal number of goals scored by the team is 1.

- (iii) Determine the median number of goals scored by the team. [1]

25 matches were played in total. As median is the middle value, we determine the number of goals scored during the 13th match (since half of 25 is 12.5, rounding to 13). At the 13th match, the median score is 2.

- (iv) Calculate the mean number of goals scored by the team. [2]

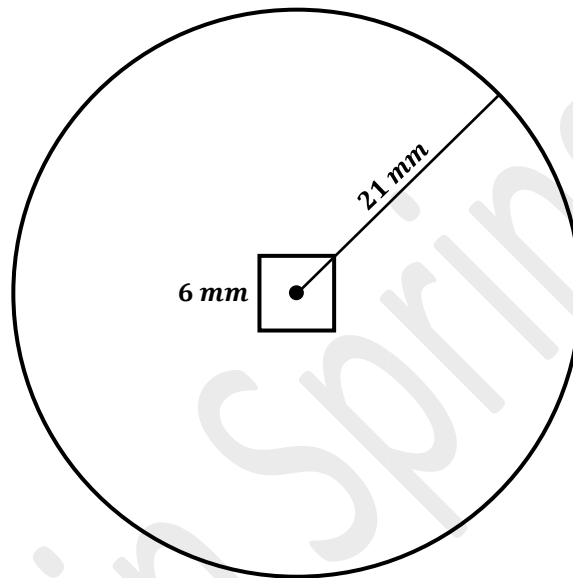
Number of matches(f)	5	7	3	3	4	2	1
Number of goals scored(x)	0	1	2	3	4	5	6
fx	0	7	6	9	16	10	6

$$\begin{aligned}
 \text{Mean} &= \frac{\sum fx}{\sum f} \\
 &= \frac{0+7+6+9+16+10+6}{5+7+3+3+4+2+1} \\
 &= \frac{54}{25} \\
 &= 2.16 \text{ goals}
 \end{aligned}$$

Total: 9 marks

6. (a) In this question, take the value of π to be $\frac{22}{7}$.

The diagram below, **not drawn to scale**, shows the cross-section of a circular metal disc of radius 21 mm. A square hole with sides 6 mm is located at the centre of the disc.



Calculate

- (i) the circumference of the disc

[1]

$$\text{Circumference of the disc} = 2\pi r$$

$$= 2 \times \frac{22}{7} \times 21$$

$$= 132 \text{ mm}$$

$\therefore$ The circumference of the disc is 132 mm.

- (ii) the area, in mm^2 , of the cross-section of the disc

$$\begin{aligned}\text{Area of whole disc} &= \pi r^2 \\ &= \frac{22}{7} \times (21)^2 \\ &= 1386 \text{ mm}^2\end{aligned}$$

$$\begin{aligned}\text{Area of square centre} &= s \times s \\ &= 6 \times 6 \\ &= 36 \text{ mm}^2\end{aligned}$$

$$\begin{aligned}\text{Area of cross-section} &= \text{Area of whole disc} - \text{Area of the square centre} \\ &= 1386 - 36 \\ &= 1350 \text{ mm}^2\end{aligned}$$

$\therefore$ The area of the cross-section of the disc is 1350 mm^2 .

- (iii) Given that the thickness of the disc is 2 mm, calculate the **maximum** number of discs that can be constructed from $1\,000\text{ cm}^3$ of available material.

$$(1\text{ cm}^3 = 1\,000\text{ mm}^3)$$

[3]

Volume of 1 disc = Cross-sectional area $\times$ thickness

$$= 1350 \times 2$$

$$= 2700\text{ mm}^3$$

Converting to cm^3 :

$$1000\text{ mm}^3 = 1\text{ cm}^3$$

$$1\text{ mm}^3 = \frac{1}{1000}\text{ cm}^3$$

$$2700\text{ mm}^3 = \frac{2700}{1000}\text{ cm}^3$$

$$= 2.7\text{ cm}^3$$

$$\text{Number of discs that can be made} = \frac{\text{Volume of metal}}{\text{Volume of 1 disc}}$$

$$= \frac{1000}{2.7}$$

$$= 370.4\text{ discs (to 1 decimal place)}$$

$\therefore$ 370 completed discs can be made.

(b) A globe is a scaled spherical representation of the earth. The actual length of the equator (LL) is 40 000 km and is represented on the globe by a piece of string of length 160 cm.

- (i) What length of string would represent an actual distance of 500 km on the globe? [1]

40 000 km is represented by 160 cm

$\therefore 1 \text{ km}$ will be represented by $\frac{160}{40\,000} \text{ cm}$

500 km will be represented by $\frac{160}{40\,000} \times 500 = 2 \text{ cm}$

- (ii) The distance between Palmyra (P) and Quintec (Q) is represented on the globe by a string of length 25 cm. Calculate the value of PQ , the actual distance, in km, between P and Q . [2]

160 cm is represented by 40 000 km

$\therefore 1 \text{ cm}$ will be represented by $\frac{40\,000}{160} \text{ km}$

25 cm will be represented by $\frac{40\,000}{160} \times 25 = 6250 \text{ km}$

$\therefore$ The distance $PQ = 6250 \text{ km}$

Total: 9 marks

7. A sequence of figures is made from squares of unit length. The first three figures in the sequence are shown below.

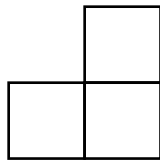


Figure 1

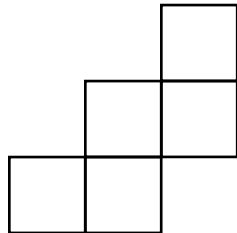


Figure 2

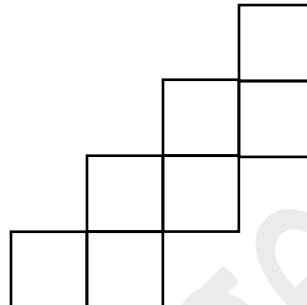


Figure 3

- (a) Draw Figure 4 of the sequence.

[2]

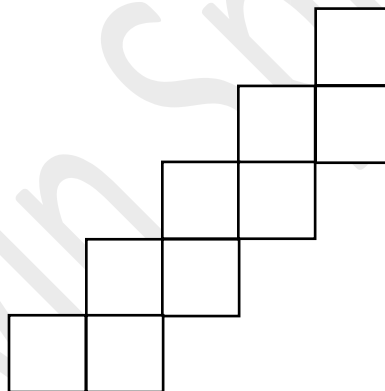


Figure 4

- (b) Study the pattern of numbers in each row of the table below. Each row relates to one of the figures in the sequence of figures on page 22. Some rows have not been included in the table.

Complete the rows numbered (i), (ii) and (iii).

	Figure	Number of Squares in Figure (S)	Perimeter of Figure (P)	
	1	3	8	
	2	5	12	
	3	7	16	
(i)	4	9	20	[2]
	$\vdots$	$\vdots$	$\vdots$	
(ii)	21	43	88	[2]
(iii)	n	$S = 2n + 1$	$P = 4n + 4$	[2]

Consider the n th term.

The number of squares, $S = 2n + 1$

The perimeter of the figure, $P = 4n + 4$

(i) When $n = 4$,

$$S = 2(4) + 1$$

$$= 8 + 1$$

$$= 9$$

$$P = 4(4) + 4$$

$$= 16 + 4$$

$$= 20$$

(ii) When $S = 43$,

$$43 = 2n + 1$$

$$2n = 43 - 1$$

$$2n = 42$$

$$n = \frac{42}{2}$$

$$n = 21$$

When $n = 21$,

$$P = 4(21) + 4$$

$$= 84 + 4$$

$$= 88$$

(c) Determine the relationship between the number of squares, S , and the perimeter, P , of a figure. [2]

$$S = 2n + 1$$

$$P = 4n + 4$$

Rearranging equation S to make n the subject of the formula gives,

$$n = \frac{S-1}{2}$$

Substituting this equation into the equation for P gives,

$$\begin{aligned}P &= 4 \left(\frac{S-1}{2} \right) + 4 \\&= 2(S-1) + 4 \\&= 2S - 2 + 4 \\&= 2S + 2\end{aligned}$$

Therefore, the relationship between the perimeter of the figure, P , and the number of squares in the figure, S , is: $P = 2S + 2$

Total: 10 marks

SECTION II

Answer ALL questions.

ALGEBRA, RELATIONS, FUNCTIONS AND GRAPHS

8. (a) The diagram on page 25 shows six points of the function $y = 3x + \frac{1}{x}$. The coordinates of these six points are given in the table below.

x	0.1	0.2	0.5	1	1.5	2	2.2	2.5
y	10.3	5.6	3.5	4	5.2	6.5	7.1	7.9

- (i) Complete the table above by calculating and inserting the missing values of y . [2]

When $x = 0.5$,

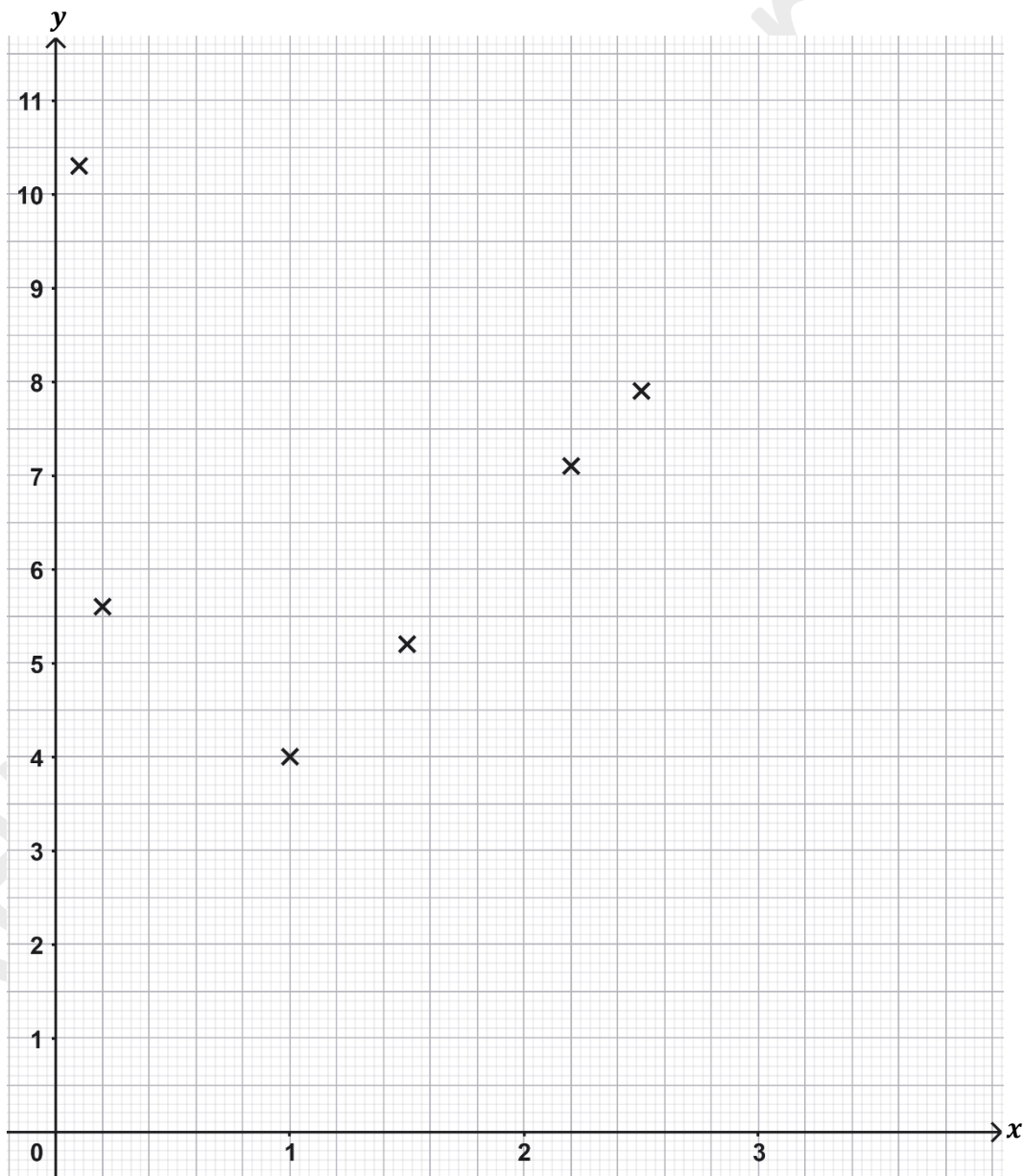
$$\begin{aligned}
 y &= 3(0.5) + \frac{1}{0.5} \\
 &= 1.5 + 2 \\
 &= 3.5
 \end{aligned}$$

When $x = 2$,

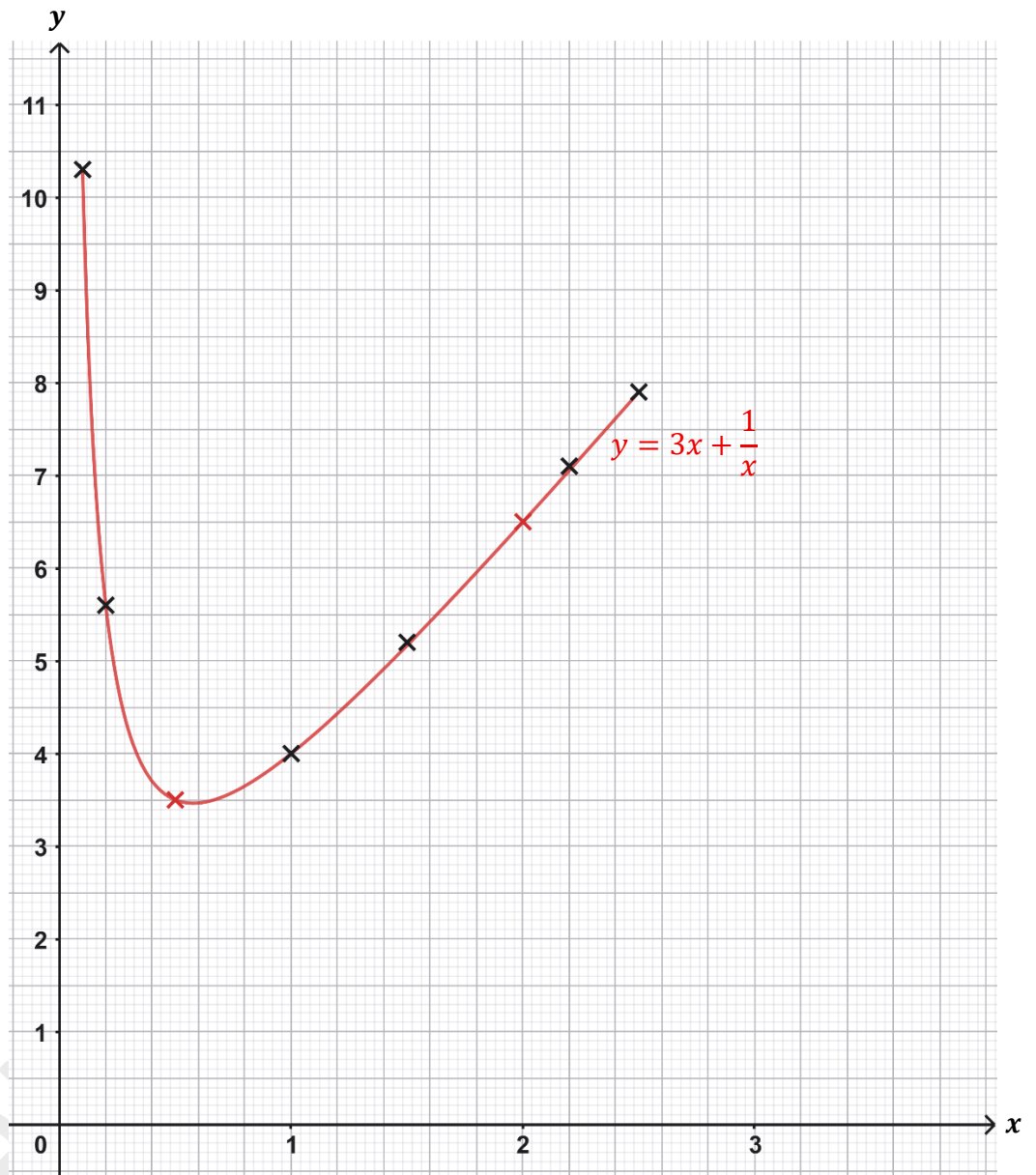
$$\begin{aligned}
 y &= 3(2) + \frac{1}{2} \\
 &= 6 + 0.5 \\
 &= 6.5
 \end{aligned}$$

Shown on the table above.

- (ii) On the diagram on page 25, the ordered pairs shown in the table have been plotted except for the missing ones. Using your answers in (a)(i), plot the missing points and connect all the points with a smooth curve.
- [2]

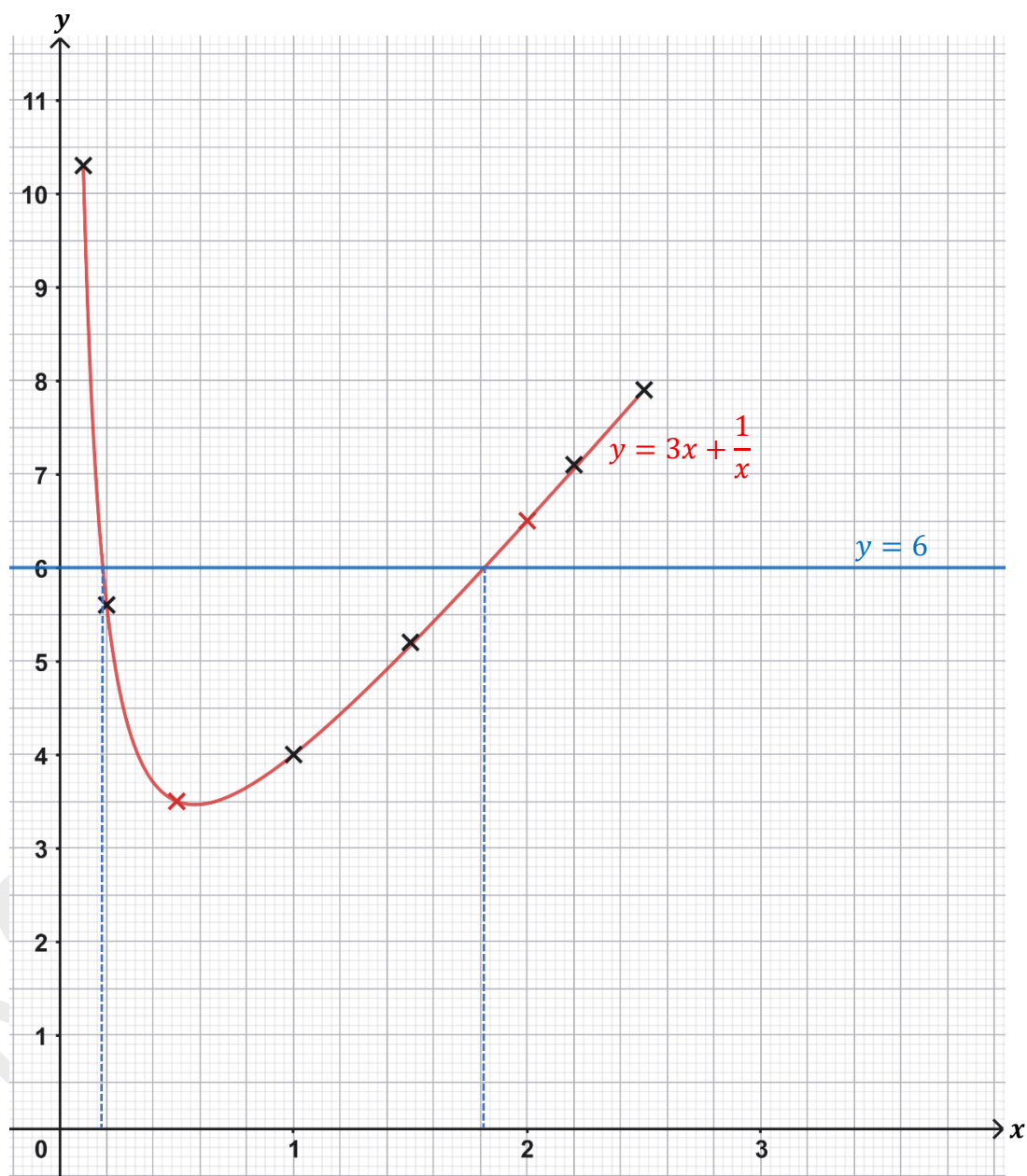


The missing values and smooth curve are shown on the next page:



- (iii) By drawing an appropriate straight line on the diagram on page 25, find approximate solutions to the equation [3]

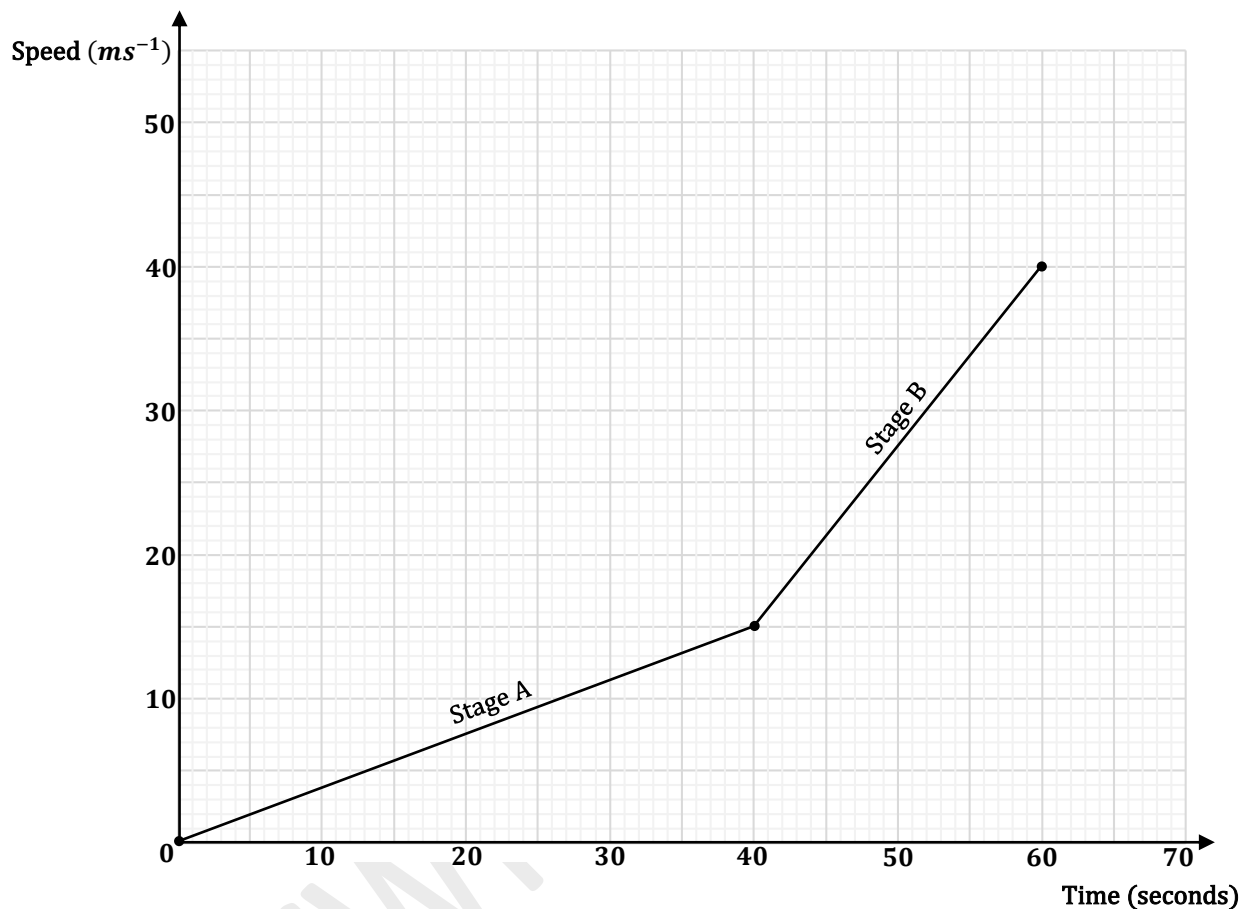
$$3x + \frac{1}{x} = 6$$



From this diagram, the two possible values of x for which

$$3x + \frac{1}{x} = 6 \text{ are } x = 0.18 \text{ and } x = 1.82.$$

- (b) The speed-time graph below shows information on the first 60 seconds of a car's journey.



- (i) Calculate the acceleration, in ms^{-2} , of the car during Stage B. [1]

To find the acceleration during stage B, we find the gradient of the graph from the beginning of stage B to the end of stage B.

First coordinate would be (40,15).

Second coordinate would be (60,40).

$$\begin{aligned}
 \text{Gradient} &= \frac{y_2 - y_1}{x_2 - x_1} \\
 &= \frac{40 - 15}{60 - 40} \\
 &= \frac{25}{20} \\
 &= 1.25 \text{ ms}^{-2}
 \end{aligned}$$

- (ii) Calculate the average speed of the car during Stage B. [3]

By finding area of the trapezium, we may find the total distance from the beginning and end of stage B.

$$\begin{aligned}
 \text{Area of trapezium} &= \frac{1}{2}(a + b)d \\
 &= \frac{1}{2}(40 + 15)(60 - 40) \\
 &= \frac{1}{2}(55)(20) \\
 &= 550 \text{ m}
 \end{aligned}$$

$$\begin{aligned}
 \text{Total time taken} &= 60 - 40 \\
 &= 20 \text{ s}
 \end{aligned}$$

$$\begin{aligned}
 \text{Average speed} &= \frac{\text{Total Distance covered}}{\text{Total time taken}} \\
 &= \frac{550}{20} \\
 &= 27.5 \text{ ms}^{-1}
 \end{aligned}$$

- (iii) At time $t = 60$ seconds, the car starts to slow down with a uniform deceleration of 2.5 ms^{-2} .

Determine how long it will take the car to come to rest. [1]

We are given that at time $t = 60$ seconds, the car starts to slow down with a uniform deceleration of 2.5 ms^{-2} .

$$\begin{aligned}\text{Gradient of the deceleration} &= \frac{40-0}{60-(60+t)} \\ &= \frac{40}{-t}\end{aligned}$$

Since the rate of deceleration is equal to the gradient, then

$$\frac{40}{-t} = -2.5$$

$$2.5t = 40$$

$$t = \frac{40}{2.5}$$

$$t = 16 \text{ s}$$

Hence, the total time to come to rest is

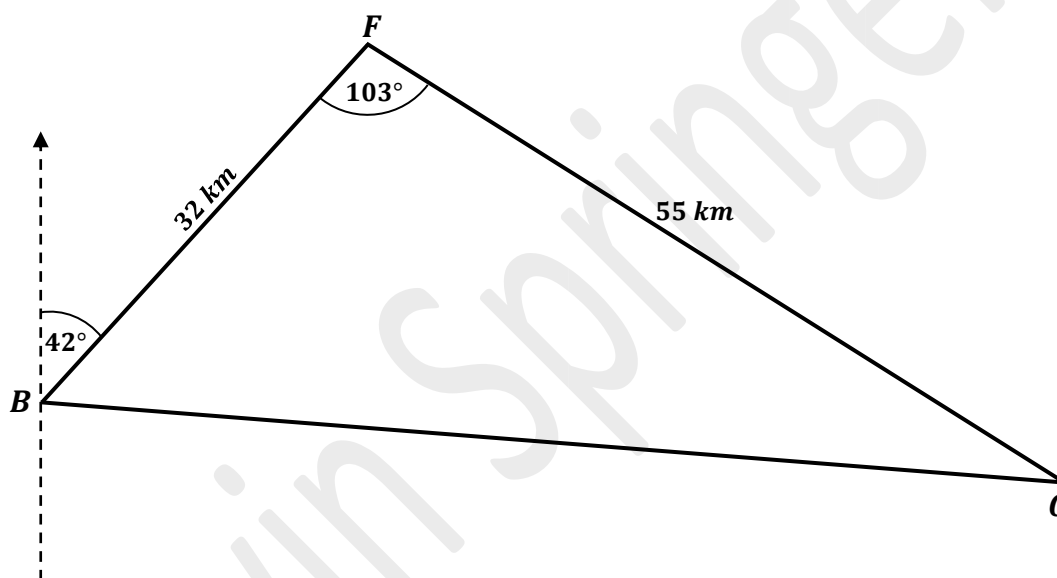
$$= 60 + 16$$

$$= 76 \text{ seconds from the start of travel.}$$

Total: 12 marks

GEOMETRY AND TRIGONOMETRY

9. (a) The diagram below, **not drawn to scale**, shows the relative positions of three reservoirs B , F and G , all on level ground. The distance $BF = 32$ km, $FG = 55$ km, $\angle BFG$ is 103° and F is on a bearing of 042° from B .



- (i) Determine the bearing of B from F .

[1]

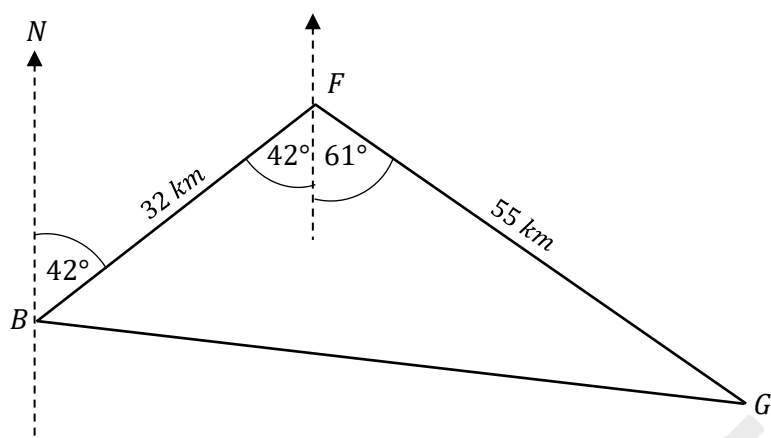
Alternate angles are equal to each other.

So, the angle made at F indicated on the diagram is 42° .

Therefore, the remaining angle is $103^\circ - 42^\circ = 61^\circ$.

The bearing of B from $F = 180^\circ + 42^\circ$

$= 222^\circ$



- (ii) Calculate the distance BG , giving your answer to one decimal place. [2]

Consider the triangle BFG .

Using the cosine rule,

$$\begin{aligned} BG^2 &= (32)^2 + (55)^2 - 2(32)(55) \cos 103^\circ \\ &= 1024 + 3025 - 3520(-0.22495) \\ &= 4049 + 791.828 \\ &= 4840.828 \end{aligned}$$

$$BG = 69.58$$

$$= 69.6 \text{ km (to 1 decimal place)}$$

- (iii) Calculate, to the nearest degree, the bearing of G from B .

Using the sine rule,

$$\frac{69.58}{\sin 103^\circ} = \frac{55}{\sin \theta}$$

$$\sin \theta = \frac{55 \times \sin 103^\circ}{69.58}$$

$$\sin \theta = 0.7702$$

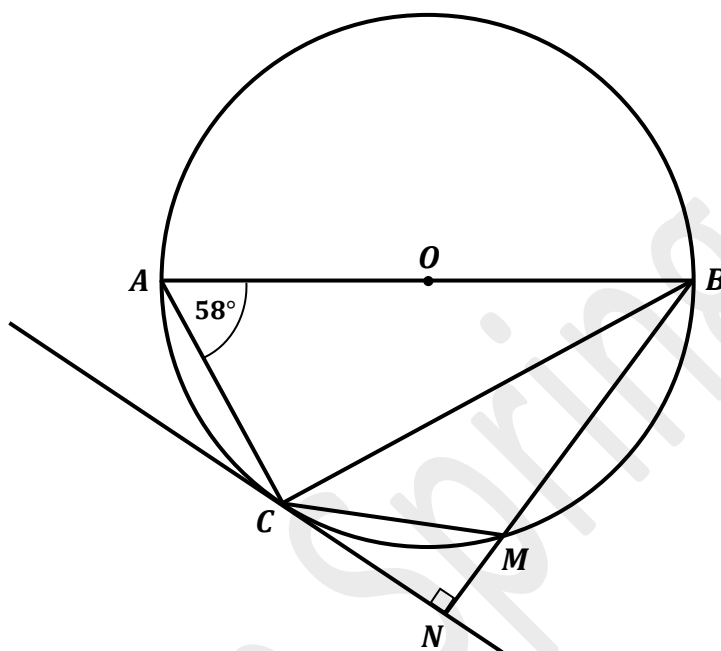
$$\theta = \sin^{-1}(0.7702)$$

$$\theta = 50.4^\circ$$

$$\begin{aligned} \text{The angle } NBG &= 42^\circ + 50.4^\circ \\ &= 92.4^\circ \end{aligned}$$

$\therefore$ The bearing of G from B is 092° (to the nearest degree).

- (b) The diagram below, **not drawn to scale**, shows a circle, with centre O . The points A, B, C and M are on the circumference. The straight line CN is a tangent to the circle at the point C and is perpendicular to BN .



Determine, giving a reason for your answer,

- (i) $\hat{ABC}$

[2]

Angle $ACB = 90^\circ$ since the angle in a semi-circle is a right angle.

The sum of the angles in a triangle add up to 180° .

Therefore,

$$\text{Angle } ABC = 180^\circ - (90^\circ + 58^\circ)$$

$$= 32^\circ$$

(ii) $\widehat{CMB}$

Since opposite angles in a cyclic quadrilateral are supplementary.

$$\text{Angle } CMB = 180^\circ - 58^\circ$$

$$= 122^\circ$$

(iii) $\widehat{NCM}$

[2]

Since angles in a straight line add up to 180° , then

$$\text{Angle } CMN = 180^\circ - 122^\circ$$

$$= 58^\circ$$

The sum of the angles in a triangle add up to 180° .

Therefore,

$$\text{Angle } NCM = 180^\circ - (90^\circ + 58^\circ)$$

$$= 32^\circ$$

Total: 12 marks

VECTORS AND MATRICES

10. (a) A transformation, T , is defined by the matrix

$$T = \begin{pmatrix} 2 & -1 \\ 2 & 0 \end{pmatrix}$$

The point $A(-2, 3)$ is mapped on to the point $A'(a, b)$ under T .

(i) Find the value of a and of b .

[2]

$$\begin{aligned} \begin{pmatrix} a \\ b \end{pmatrix} &= \begin{pmatrix} 2 & -1 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} -2 \\ 3 \end{pmatrix} \\ &= \begin{pmatrix} (2 \times -2) + (-1 \times 3) \\ (2 \times -2) + (0 \times 3) \end{pmatrix} \\ &= \begin{pmatrix} -4 - 3 \\ -4 + 0 \end{pmatrix} \\ &= \begin{pmatrix} -7 \\ -4 \end{pmatrix} \end{aligned}$$

$$\therefore a = -7 \text{ and } b = -4$$

- (ii) Determine the transformation matrix that maps A' to A .

$$T = \begin{pmatrix} 2 & -1 \\ 2 & 0 \end{pmatrix}$$

$$\det(T) = ad - bc$$

$$= (2)(0) - (-1)(2)$$

$$= 0 + 2$$

$$= 2$$

$$\text{adj}(T) = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 1 \\ -2 & 2 \end{pmatrix}$$

$$\therefore T^{-1} = \frac{1}{\det} \times \text{adj}(T)$$

$$= \frac{1}{2} \begin{pmatrix} 0 & 1 \\ -2 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & \frac{1}{2} \\ -2 & \frac{2}{2} \end{pmatrix}$$

$$= \begin{pmatrix} 0 & \frac{1}{2} \\ -1 & 1 \end{pmatrix}$$

(iii) Another transformation, P , is defined by the matrix $P = \begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix}$.

(a) Find the single 2×2 matrix that represents the combined

transformation of T followed by P .

[2]

$$P = \begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix}$$

The combined transformation T followed by P is,

$$\begin{aligned} PT &= \begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 2 & 0 \end{pmatrix} \\ &= \begin{pmatrix} (0 \times 2) + (1 \times 2) & (0 \times -1) + (1 \times 0) \\ (1 \times 2) + (-2 \times 2) & (1 \times -1) + (-2 \times 0) \end{pmatrix} \\ &= \begin{pmatrix} 0 + 2 & 0 + 0 \\ 2 + (-4) & -1 + 0 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 0 \\ -2 & -1 \end{pmatrix} \end{aligned}$$

(b) Hence, find the image of the point $(1, 4)$ under this combined

transformation.

[1]

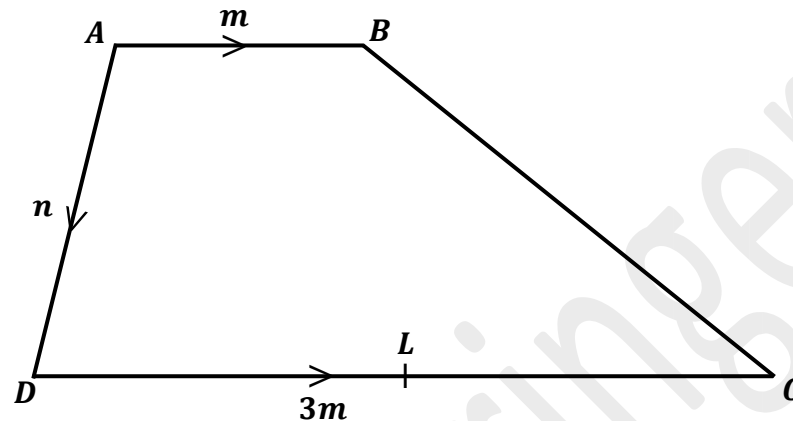
The image of the point $(1, 4)$ under this transformation is

$$\begin{aligned} \begin{pmatrix} 2 & 0 \\ -2 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 4 \end{pmatrix} &= \begin{pmatrix} (2 \times 1) + (0 \times 4) \\ (-2 \times 1) + (-1 \times 4) \end{pmatrix} \\ &= \begin{pmatrix} 2 + 0 \\ -2 - 4 \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ -6 \end{pmatrix} \end{aligned}$$

$\therefore$ The image is $(2, -6)$.

(b) The diagram below, **not drawn to scale**, shows a quadrilateral $ABCD$ in which

$$\overrightarrow{AB} = \mathbf{m}, \quad \overrightarrow{DC} = 3\mathbf{m} \quad \text{and} \quad \overrightarrow{AD} = \mathbf{n}.$$



- (i) Complete the statement below on the geometric properties of the following vectors.

$\overrightarrow{AB}$ and $\overrightarrow{DC}$ are parallel and

$|\overrightarrow{AB}|$ is $\frac{1}{3}$ times $|\overrightarrow{DC}|$. [2]

Since $\overrightarrow{AB}$ and $\overrightarrow{DC}$ are parallel, they are scalar multiples of each other.

In fact, $\overrightarrow{DC}$ is larger than $\overrightarrow{AB}$ by a scalar factor of 3.

Since $\overrightarrow{DC} = 3\overrightarrow{AB}$, then $|\overrightarrow{AB}| = \frac{1}{3}|\overrightarrow{DC}|$.

Therefore, the completed statement would be:

$\overrightarrow{AB}$ and $\overrightarrow{DC}$ are parallel and $|\overrightarrow{AB}|$ is $\frac{1}{3}$ times $|\overrightarrow{DC}|$.

- (ii) Express $\overrightarrow{BC}$ in terms of m and n .

According to the Triangle Law,

$$\begin{aligned}\overrightarrow{BC} &= \overrightarrow{BA} + \overrightarrow{AC} \\ &= -m + \overrightarrow{AC}\end{aligned}$$

$$\begin{aligned}\text{Now, } \overrightarrow{AC} &= \overrightarrow{AD} + \overrightarrow{DC} \\ &= n + 3m\end{aligned}$$

$$\begin{aligned}\text{So, } \overrightarrow{BC} &= -m + (n + 3m) \\ &= n + 2m\end{aligned}$$

- (iii) L is the midpoint of $\overrightarrow{CD}$. Find $\overrightarrow{BL}$ in terms of m and n . [2]

$$\begin{aligned}\overrightarrow{AL} &= \overrightarrow{AD} + \overrightarrow{DL} \\ &= \overrightarrow{AD} + \frac{1}{2}\overrightarrow{DC} \\ &= n + \frac{1}{2}(3m)\end{aligned}$$

So,

$$\begin{aligned}\overrightarrow{BL} &= \overrightarrow{BA} + \overrightarrow{AL} \\ &= -m + n + \frac{1}{2}(3m)\end{aligned}$$

$$= \frac{1}{2}m + n$$

Total: 12 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.